

Balance challenge cards (predict first)

Four indoor challenges that replace a real ropes course. Do EVERY challenge next to a wall so a slip just leaves you leaning. Keep the floor clear; a helper spots the chair stand; the taped line stays flat (no running, no raised beam). Print the card page once per team; mark WORK or FAIL on the card FIRST, then test, then map the forces. The reasons are on the instructor answer key so the prediction stays a real prediction.

1 Wall and Heel

Stand with your back and heels flat against a wall. Keeping them touching, try to reach a coin placed on the floor a little ahead of you, or lift one foot forward off the floor.

My prediction — will it WORK or FAIL? _____ Where does my center of mass go? _____

2 Backless Chair Stand

Sit tall on a backless chair, feet flat, arms crossed. Try to stand up WITHOUT leaning your chest forward.

My prediction — will it WORK or FAIL? _____ Where does my center of mass go? _____

3 Toe Grab Hop

Bend down, grab your toes, and try to hop forward while still holding them.

My prediction — will it WORK or FAIL? _____ Where does my center of mass go? _____

4 Loaded Line Walk

Walk a taped floor line holding a small weight straight out at arm's length, then again holding it at your chest. Which is steadier?

My prediction — will it WORK or FAIL? _____ Where does my center of mass go? _____

Instructor answer key (do not hand out)

Keep this page with the instructor and print it once. Reveal a challenge's reason only after each team has recorded its prediction.

#	CHALLENGE	RESULT	CENTER-OF-MASS REASON
1	Wall and Heel	Fails	The wall blocks your hips from shifting back to counterbalance, so as you reach forward your center of mass passes beyond your toes, out of your base of support, and you tip.
2	Backless Chair Stand	Fails without a lean	To stand you must first move your center of mass forward over your feet. With no chair back and no forward lean, it stays behind your feet, so you cannot rise.
3	Toe Grab Hop	Very hard	Holding your toes locks your center of mass low and forward, so you cannot swing your base under it to launch a hop.
4	Loaded Line Walk	Steadier at the chest	At arm's length the weight pulls your center of mass forward and sideways off the line; held at the chest it stays over your base, so you wobble less.